STA421: Foundations of Bayesian Methodology | UZH 2024

Individual project

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0 Code initialisation

```r
rm(list = ls())

library(rjags)
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.2
```

```
## Loaded modules: basemod,bugs
```

```r
library(MASS)
library(glue)
library(rjags)
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.7.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Zurich
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] glue_1.8.0    MASS_7.3-61   rjags_4-17    coda_0.19-4.1
##
## loaded via a namespace (and not attached):
##  [1] digest_0.6.37     fastmap_1.2.0     xfun_0.52         lattice_0.22-6
##  [5] knitr_1.50        htmltools_0.5.8.1 rmarkdown_2.29    cli_3.6.4
##  [9] grid_4.4.2        compiler_4.4.2    rstudioapi_0.17.1 tools_4.4.2
## [13] evaluate_1.0.3    yaml_2.3.10       rlang_1.1.5
```

1 Aim of the study

The aim of the study is a proof-of-concept to assess the efficacy and safety of the anti-IL-17A monoclonal antibody "Secukinumab" in patients with active ankylosing spondylitis, which is an inflammatory disease that affects the spine. Secukinumab is expected to reduce the inflammatory response mediated by IL-17A and thus to improve the course of disease. The goal of the study is ultimately to help path the way to an alternative therapy for patients for whom current treatments are not working. The following analysis provides guidance on optimal study design, sample sizes, elicitation of priors for placebo and treatment groups, statistical analysis and interpretation of the results.

2 Study design and sample size computation

2.1 Elicitation of Placebo prior distribution

Based on historical data from 8 randomized ankylosing spondylitis studies, we conduct a random effects meta analysis to elicitate the predictive distribution of the proportion of responders on placebo in a new study.

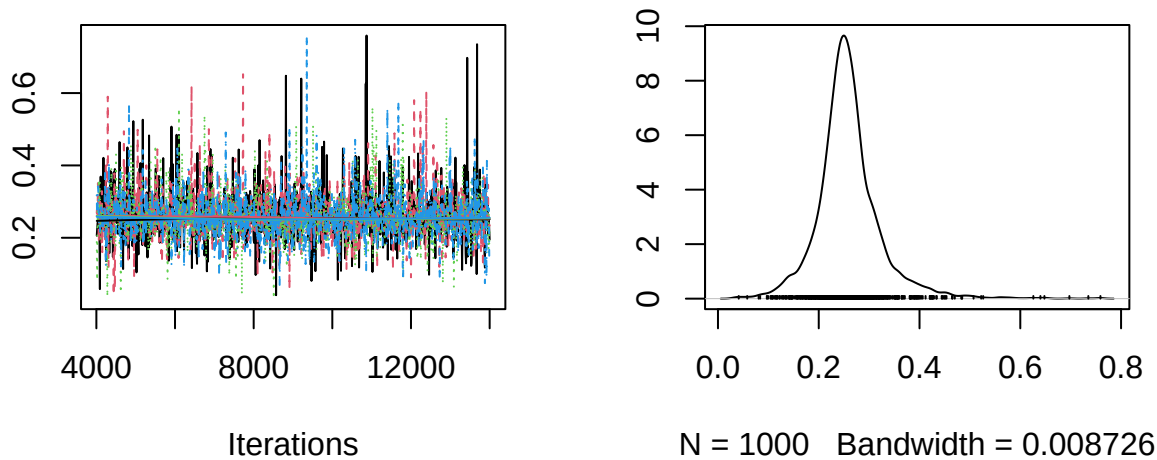
```r
# Defining colors
cols <- c(rgb(red = 0.4, green = 0, blue = 0.3, alpha = 0.5),  # placebo
          rgb(red = 0, green = 0.5, blue = 0.3, alpha = 0.3))  # treatment

cols_opaque <- c("#b27fa5", # placebo
                 "#b2d9c9") # treatment
```

```r
# Assume that the data for 8 historical studies subject to placebo are as follows:
# Total number of observations ni in each study subject to placebo i = 1, ... , 8 :
pl_total <- c(107,44,51,39,139,20,78,35)
# Number of cases xi in each study subject to placebo i = 1, ... , 8:
pl_case <- c(23,12,19,9,39,6,9,10)
# Log odds:
pl_logit <- log( pl_case / (pl_total - pl_case) )
# SE:
pl_se <- sqrt( 1/pl_case + 1/(pl_total - pl_case))

# JAGS model
pl1_modelString = "
model {
for(i in 1:length(y)) {
  y[i] ~ dnorm(theta[i], prec_s[i]);
  theta[i] ~ dnorm(mu, prec_tau);
}
theta_new ~ dnorm(mu, prec_tau); # predictive distribution for theta
p_new <- exp(theta_new)/(1+exp(theta_new)); # predictive distribution at the probability scale
mu ~ dnorm(0.0, 1.0E-4);
prec_tau ~ dgamma(1.0E-3, 1.0E-3); # just our assumption
}
```

```r
"
writeLines(pl1_modelString, con="pl1_model.txt") # write to a file
pl_data <- list(y = pl_logit, prec_s = 1/pl_se^2 )

# prec_tau: Expectation of dgamma(1.0E-3, 1.0E-3) = 1.0E-3 / 1.0E-3 = 1
pl_inits <- list(
  list(mu=0, prec_tau = 1, .RNG.name = "base::Wichmann-Hill", .RNG.seed = 111),
  list(mu=0, prec_tau = 1, .RNG.name = "base::Marsaglia-Multicarry", .RNG.seed = 222),
  list(mu=0, prec_tau = 1, .RNG.name = "base::Super-Duper", .RNG.seed = 333),
  list(mu=0, prec_tau = 1, .RNG.name = "base::Mersenne-Twister", .RNG.seed = 444))

# model initiation
model.jags <- jags.model(
  file = "pl1_model.txt",
  data = pl_data,
  inits = pl_inits,
  n.chains = 4,
  n.adapt = 4000)
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 8
##    Unobserved stochastic nodes: 11
##    Total graph size: 34
##
## Initializing model
```

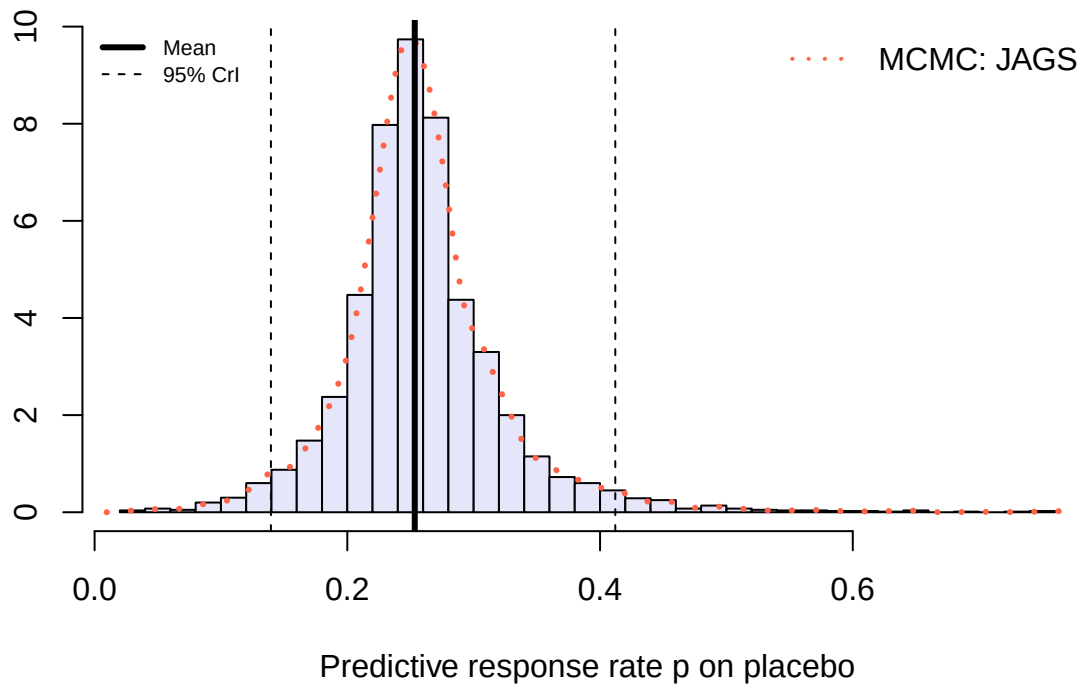
```r
# burn-in
update(model.jags, n.iter = 4000)

# sampling
fit.jags.coda <- coda.samples(
  model = model.jags,
  variable.names = c("p_new"),
  n.iter = 10000,
  thin = 10
)

#summary(fit.jags.coda)
par(mfrow=c(1,2), mar=c(10.1, 3.1, 5, 2.1))
plot(fit.jags.coda, main = "")
```




```r
m.fit.jags.coda <- as.matrix(fit.jags.coda)
p_new.sim <- m.fit.jags.coda[, "p_new"]

glue("Summary of posterior predictive distribution for p:")
glue("Mean alpha: {formatC(mean(p_new.sim), 5)} ")
glue("SD alpha: {formatC(sd(p_new.sim), 5)} ")
glue("Quantiles:")
quantile(p_new.sim, c(0.025, 0.5, 0.975) )
```

```
## Summary of posterior predictive distribution for p:
## Mean alpha: 0.25915
## SD alpha: 0.064764
## Quantiles:
##      2.5%       50%     97.5%
## 0.1395257 0.2533616 0.4120539
```

```r
# Histogram and density of simulated p's
truehist(p_new.sim, prob=TRUE, col="lavender", xlab="Predictive response rate p on placebo")
lines(density(p_new.sim),lty=3,lwd=3, col="tomato")
legend("topright",c("MCMC: JAGS"),lty=3,lwd=2,col="tomato",cex=1.0,bty="n")
abline(v = median(p_new.sim), lwd = 3, lty = 1)
abline(v = quantile(p_new.sim, c(0.025, 0.975) ), lty = 2 )
legend("topleft", legend = c("Mean", "95% CrI"), lty = c(1,2), lwd = c(3,1), bty = "n", cex = 0.7)
```



The model is a Bayesian hierarchical model designed to estimate the overall log-odds of response in the placebo group across multiple studies, accounting for between-study variability. It uses non-informative priors to ensure that primarily the data inform the posterior distributions. The predictive distribution is an estimation of the response probability for a new, hypothetical study. We obtain a predictive response rate of 25.34% for placebo with a Credible Interval of 13.95% - 41.21%. The Credible Interval means that the response rate lies in this range with 95% probability.

2.2 Sample size calculation

Classical approach

```
samplesize <- power.prop.test(p1 = 0.60, p2 = 0.25, power = 0.8, sig.level = 0.05)$n
```

To achieve a power of 80% at a significance level of 5% with a classical approach, assuming a placebo response ratio of 25% and a treatment response ratio of 60%, the study would require 31 participants per group, i.e. 62 patients in total. The drawback of a non-bayesian approach is that many patients need to be allocated to the placebo group, who then cannot benefit from the treatment.

We therefore suggest to opt for a bayesian approach that leverages historic patient data. We suggest a 4:1 study design to allow many patients to benefit from the treatment.

We now want to find the minimum required number of patients per group for the study, given the following:

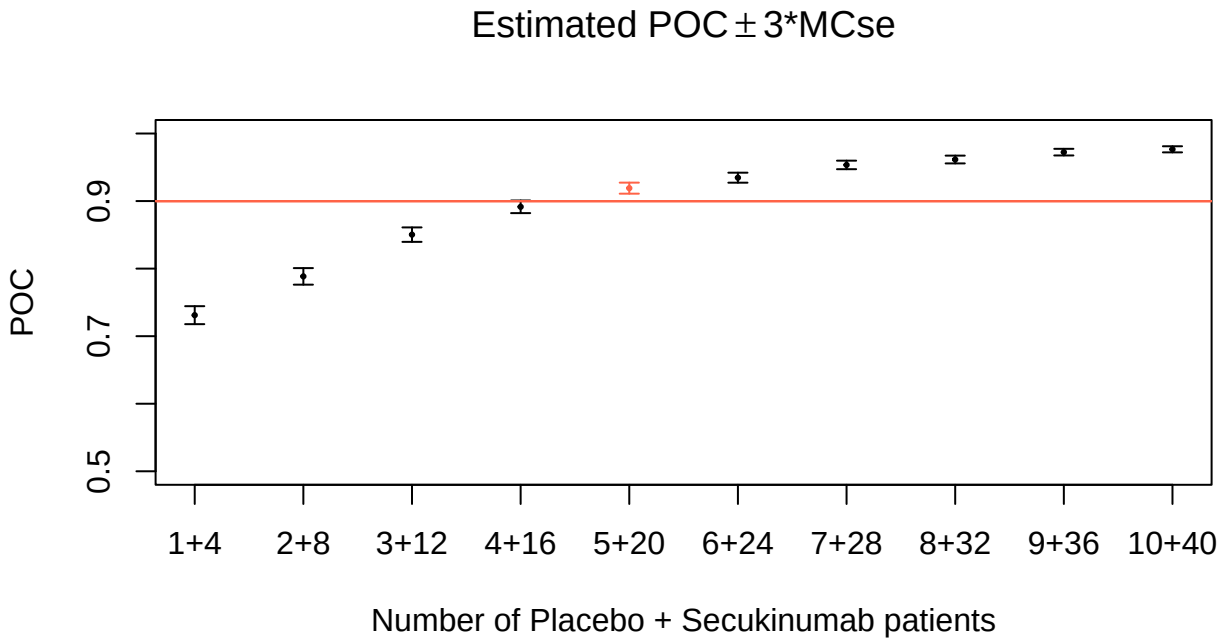
- We assume that the true ASAS20 response rate on Secukinumab is 60%, and on placebo it's 25% (based on the meta analysis)
- The ratio of Secukinumab patients to placebo patients should be 4:1, for example 40 Secukinumab and 10 placebo patients, allowing more patients to benefit from the treatment
- The proof of concept (POC) requires that the response rate on Secukinumab is larger than on placebo
- We want to achieve a POC > 90%

The process is as follows:

For a given number of Secukinumab and placebo patients in a ratio of 4:1, we simulate 10'000 times the number of responders in each group, under our hypothesis that the true response rates are 60% (Secukinumab) and 25% (placebo). Dividing the number of responders by the number of patients in the corresponding group yields 10'000 proportions of responders for each group. The ratio of simulations where the proportion of responders was higher for Secukinumab than for Placebo is the estimate for our Proof of Concept (POC). We also calculate the standard deviation of this POC estimate, which is called the Monte Carlo standard error (MCse), to know how much uncertainty there is around this estimation. If we still don't undershoot our 90% threshold after subtracting 3 times the standard deviation from the POC estimate, the probability of the POC being lower than 90% can be regarded as good as zero.

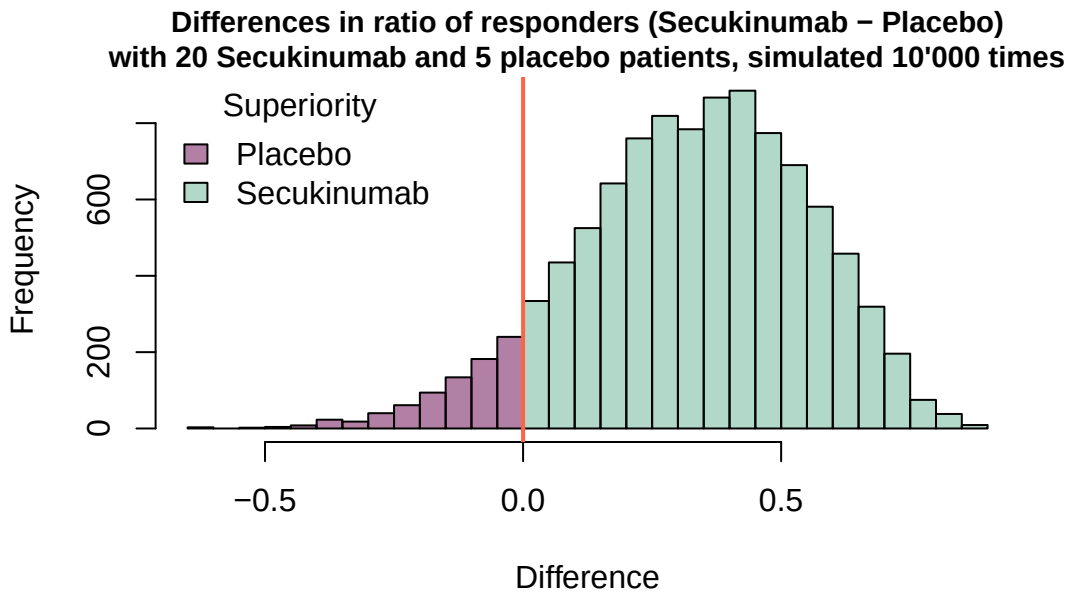
```r
# Set seed
set.seed(123456789)

# Hypotheses for true response rates
hypothesis_rr_S <- 0.6
hypothesis_rr_P <- 0.25

# Potential range for number of placebo patients
# Starting with smallest possible value 1
n_P_range <- seq(1,10)

# Initiate plot (Number of patients vs. POC)
par(mar = c(8,4,5,1))
plot(n_P_range, rep(NA, length(n_P_range) ), ylim = c(0.5,1),
     main = expression("Estimated POC" %+-% "3*MCse"), xaxt = 'n',
     xlab = "Number of Placebo + Secukinumab patients", ylab = "POC")
# Label x-axis ticks with number of Placebo and Secukinumab patients
axis(1, at = n_P_range, labels = paste0( n_P_range, "+", 4*n_P_range  ) )

# We have not found the minimum required number of patients yet
found_min_n <- 0

# Number of simulations
M <-10000

# Run simulation for a given candidate number of Placebo patients
for (n_P_sim in n_P_range){

  set.seed(123456789)
  color <- "black" # Default plotting color

  # Number of Secukinumab patients = 4 * number of Placebo patients
  n_S_sim <- 4*n_P_sim
```

```r
  # Sampling the number of responders from binomial distribution with hypothesised p
  # And dividing by the number of patients --> ratio of responders (2 vectors of size M)
  rr_S <- rbinom(M, size = n_S_sim, prob = hypothesis_rr_S) /n_S_sim
  rr_P <- rbinom(M, size = n_P_sim, prob = hypothesis_rr_P) /n_P_sim

  # POC and MCse
  indicator <- as.numeric(rr_S > rr_P) # indicating if rr_S is superior to rr_P
  POC <- sum(indicator) / M # = mean(indicator)
  MCse <- 1/sqrt(M) * sqrt( sum((indicator - POC )^2 / (M-1) )) # Monte Carlo standard error

  # Condition: Only if it's the first n that achieves POC - 3 * MCse > 90%:
  if (POC - 3 * MCse > 0.9 & found_min_n == 0){

    found_min_n <- 1 # We now have found the minimum n
    color <- "tomato" # In this case, plot POC in red

    # Store values for the report later
    differences_min <- rr_S - rr_P
    n_P_min <- n_P_sim
    n_S_min <- n_S_sim
    POC_min <- POC
    MCse_min <- MCse
  }

  # Plot POC for the given number of patients
  points(n_P_sim, POC, pch = 19, cex = 0.3, col = color )
  # Errorbar for plus/minus 3*MCse
  arrows(n_P_sim, POC- 3*MCse, n_P_sim, POC+ 3*MCse, length=0.05, angle=90, code=3, col = color )
  # Threshold
  abline(h = 0.9, col = "tomato")
}
```




```
## Min. number of Placebo patients required: 5
## Min. number of Secukinumab patients required: 20
## Estimated POC: 0.9191
## Estimated MCse(POC): 0.0027
```

```r
# Plot histogram of differences
par(mar = c(5,4,2,5), oma = c(0,0,6,1))
breaks <- pretty(differences_min, 25)
col <- ifelse( breaks < 0, cols[1], cols[2])
hist(differences_min, xlab = "Difference",
main = "Differences in ratio of responders (Secukinumab - Placebo)\nwith 20 Secukinumab and 5 placebo pa
abline(v = 0, lwd = 2, col = "tomato")
legend("topleft", legend = c("Placebo", "Secukinumab"),title = "Superiority", fill = cols, bty = "n")
```



We applied the above process to different patient group sizes that follow a 4:1 ratio, and determined the smallest group sizes that achieved a Proof of Concept (POC) greater than 90% after subtracting 3 times the MCse. We conclude that the study requires 5 placebo and 20 Secukinumab patients, i.e. a total of 25 patients, which will result in an estimated POC of 0.9191 with an MCse of 0.0027, under the assumption of our hypothesis about the true response rates.

Group	Assumed Response Rate	Min. required n	Est. POC	MCse(POC)
Secukinumab	60%	20	0.9191	0.0027
Placebo	25%	5		

3 Priori elicitation for placebo and treatment groups

3.1 Placebo

The predictive distribution of the placebo response rate that was derived from the meta analysis of historical data is now approximated by a Beta distribution. The parameters α and β are chosen such that they match the mean and variance of the predictive distribution, using the following formulas:

$$\alpha = \frac{E(X)[-E(X)^2 - Var(X) + E(X)]}{Var(X)}$$
$$\beta = \frac{[E(X) - 1][E(X)^2 + Var(X) - E(X)]}{Var(X)}$$

```
beta_params <- function(mean, var){

  # Input:
  # mean: Sample mean of a beta distribution
  # var: Sample variance of a beta distribution

  alpha <- (mean * (-mean^2 - var + mean)) / var
  beta <- ((mean - 1) * (mean^2 + var - mean)) / var

  # Output:
  # alpha: alpha parameter for the beta distribution
  # beta: beta parameter for the beta distribution
  # returned as a list

  params <- list(alpha, beta)
  return(params)
}

params <- beta_params(mean(p_new.sim), var(p_new.sim))
```

Group	n studies	Est. response rate	95% CrI	Est. α	Est. β
Placebo	8	25.92%	13.95 - 41.21%	12	33

The elicitated parameters for the placebo prior beta distribution are alpha = 12 and beta = 33.

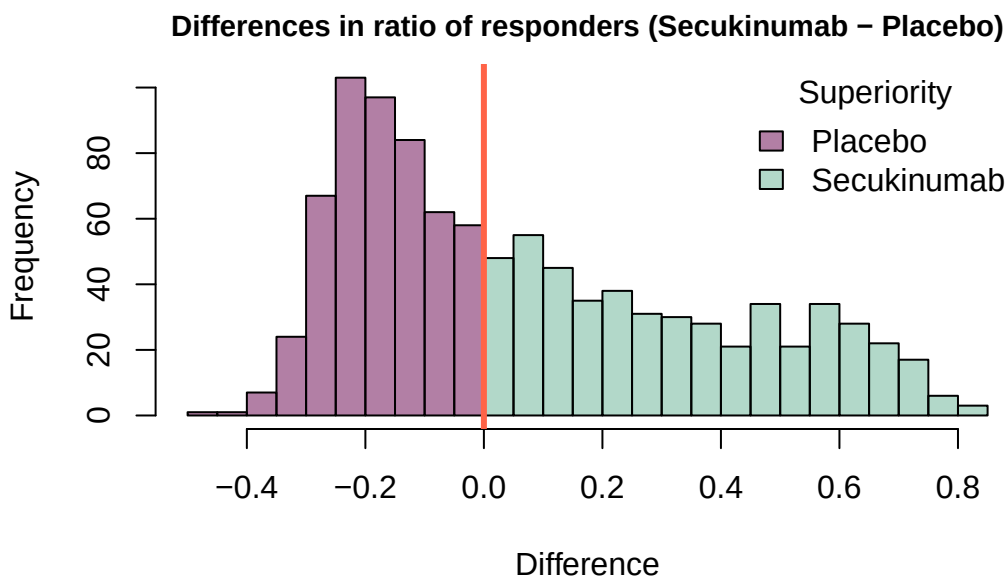
3.2 Treatment

A beta distribution assigns probabilities to probabilities, or in this case to response rates. For the treatment group, we want to choose a prior Beta distribution with parameters α and β such that there is around 50% chance that the treatment response rate is superior, and 50% chance that the placebo response rate is superior, based on the prior distributions only.

We don't want to put any premature confidence into the treatment effect, and let the data speak for itself. This is good scientific practice. To obtain a sceptical beta distribution that puts a lot of weight on low response rates, we fix the second shape parameter (β) of the treatment's prior to 1, and propose 0.5 for the first shape parameter (α) to fulfill the criterion of equal superiority.

```
# Placebo
prior_shape1_P <- 11
prior_shape2_P <- 32

# Treatment
prior_shape2_S <- 1 # Fixing 1 for the 2nd parameter
```

```r
prior_shape1_S <- 0.5 # Proposing 0.5 for the 1st parameter

M <- 1000

# Drawing 1000 response rate samples each from beta distributions
p_P <- rbeta(M, prior_shape1_P, prior_shape2_P)
p_S <- rbeta(M, prior_shape1_S, prior_shape2_S)

diffs <- p_S - p_P # Difference treatment - placebo response rates

indicator <- as.numeric(diffs > 0) # indicating if p_S is superior to p_P
mean_superior_S <- mean(indicator) # ratio indicating how often this is the case

MCse <- 1/sqrt(M) * sqrt( sum((indicator - mean_superior_S)^2 / (M-1) ))

# Histogram
par(mar = c(5,4,2,5), oma = c(0,0,6,1))
breaks <- pretty(diffs, 25)
col <- ifelse( breaks < 0, cols[1], cols[2])

hist(diffs, xlab = "Difference",
     main = "Differences in ratio of responders (Secukinumab - Placebo)",
     breaks = breaks, col = col, cex.main = 0.9)
abline(v = 0, lwd = 3, col = "tomato")
legend("topright", legend = c("Placebo", "Secukinumab"),
       title = "Superiority", fill = cols, bty = "n")
```

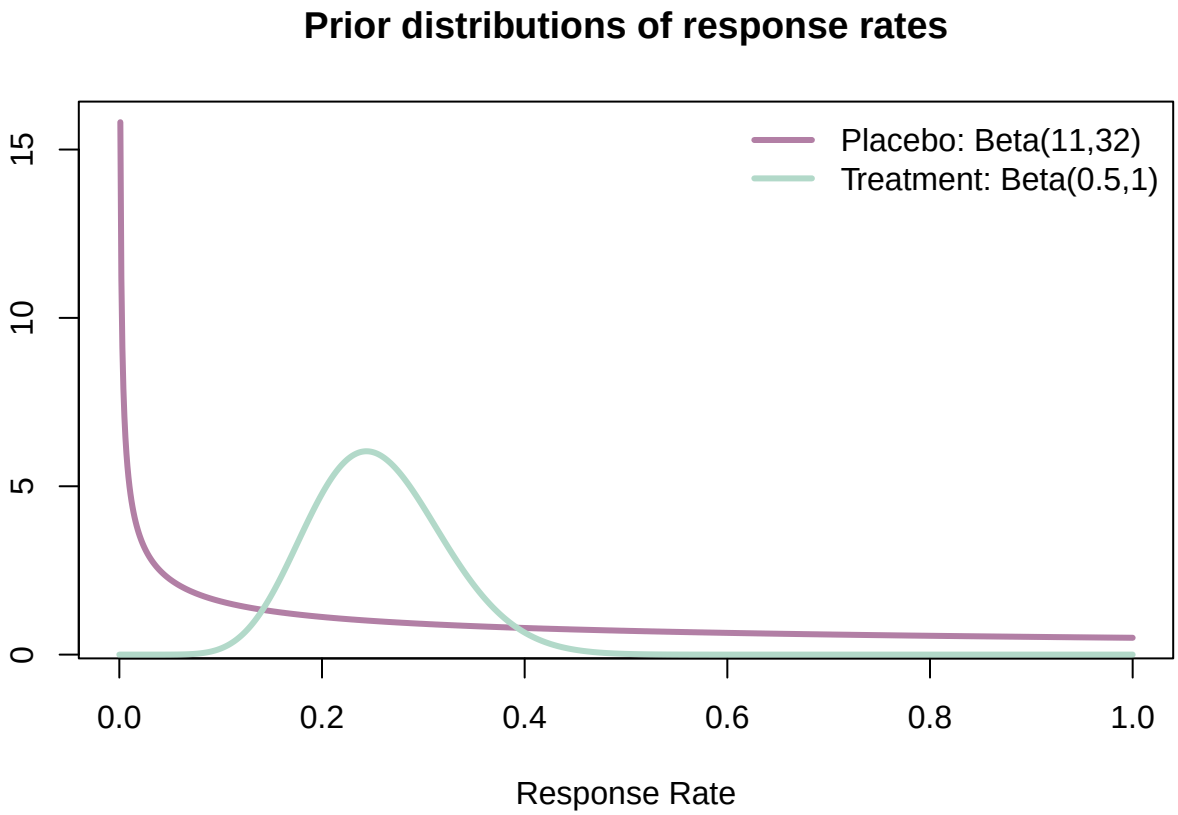


To show that (0.5,1) are indeed appropriate shape parameters for the prior distribution of the treatment response rate, we sample 1000 times from the treatment prior Beta(0.5,1), and 1000 times from the placebo prior Beta(11,32). This results in 1000 response rates per group. We then check how often the response rate was higher in the treatment group than in the placebo group. This turned out to be the case in 49.6% of the simulations, and the Monte Carlo standard error for this estimation is only 0.0158188, which means that the estimate is quite precise. This shows that the parameters (0.5,1) are appropriate to assume equal superiority of response rates, based on the prior distributions.

We will use the prior distribution Beta(11,32) for placebo and the prior distribution Beta(0.5,1) for the treatment.

```
## Prior for Secukinumab: Beta(0.5,1)
## Mean: 0.3333
## Median: 0.2500
## 95% CrI: [0.0006, 0.9506]
```

```
## Prior for placebo: Beta(11,32)
## Mean: 0.2558
## Median: 0.2520
## 95% CrI: [0.1386, 0.3945]
```



The two distributions have approximately the same median. The prior for the treatment Beta(0.5, 1) is a skeptical prior and has a very wide 95% credible interval, while the prior for placebo Beta(11, 32) is more precise and accounts for 11+32=43 observations, which is more than the sample size of the treatment group, so it's an enthusiastic prior with a much narrower 95% credible interval.

4 Data collection

The number of patients participating in the study is 23 on Secukinumab, and 6 on placebo, which fulfills the minimal sample size criteria.

The following data was collected:

Group	n	Responders x (%)
Secukinumab	23	14 (60.9%)
Placebo	6	1 (16.7%)

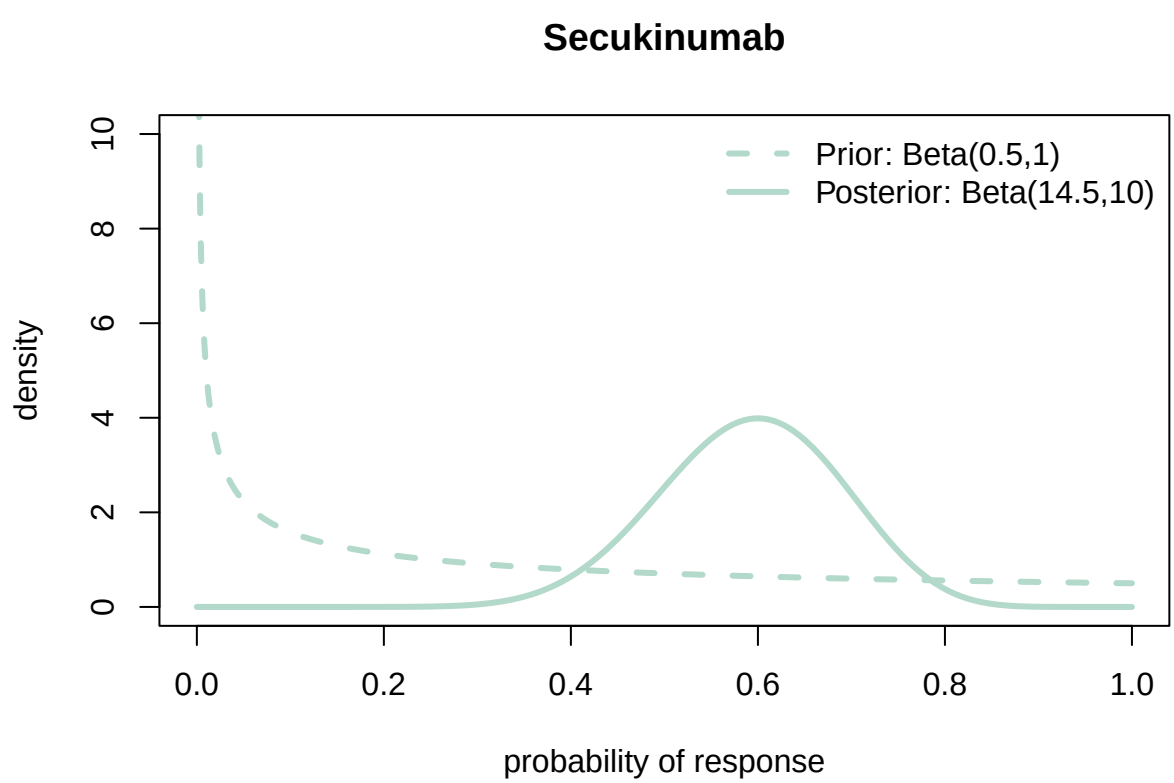
5 Data analysis

After obtaining the data, we can now calculate the posterior distributions of the response rates by combining the prior distributions with the data.

The posterior $p|y$ follows a $Beta(\alpha + n\overline{y},\ \beta + n - n\overline{y})$ distribution, where $n\overline{y}$ is the number of responders in the corresponding group. The posterior alpha parameter is calculated by adding the number of observed responders, and the new beta parameter is calculated by adding the number of observed non-responders.

5.1 Secukinumab posterior distribution

```
n_ybar_S <- 14 # number of responders
n_S <- 23
post_shape1_S <- prior_shape1_S + n_ybar_S
post_shape2_S <- prior_shape2_S + n_S - n_ybar_S
x <- seq(0,1, by = 0.001)

plot(x, dbeta(x, prior_shape1_S, prior_shape2_S), col = cols_opaque[2], lwd = 3, lty = 2,
     type = "l", ylab = "density", xlab = "probability of response",
     main = "Secukinumab", ylim = c(0,10) )
lines( x, dbeta(x, post_shape1_S, post_shape2_S), col = cols_opaque[2], lwd = 3 )
legend("topright", legend = c(sprintf("Prior: Beta(%g,%g)", prior_shape1_S, prior_shape2_S),
                              sprintf("Posterior: Beta(%g,%g)", post_shape1_S, post_shape2_S)),
       col = cols_opaque[2], lty = c(2,1), lwd = 3, bty = "n")
```



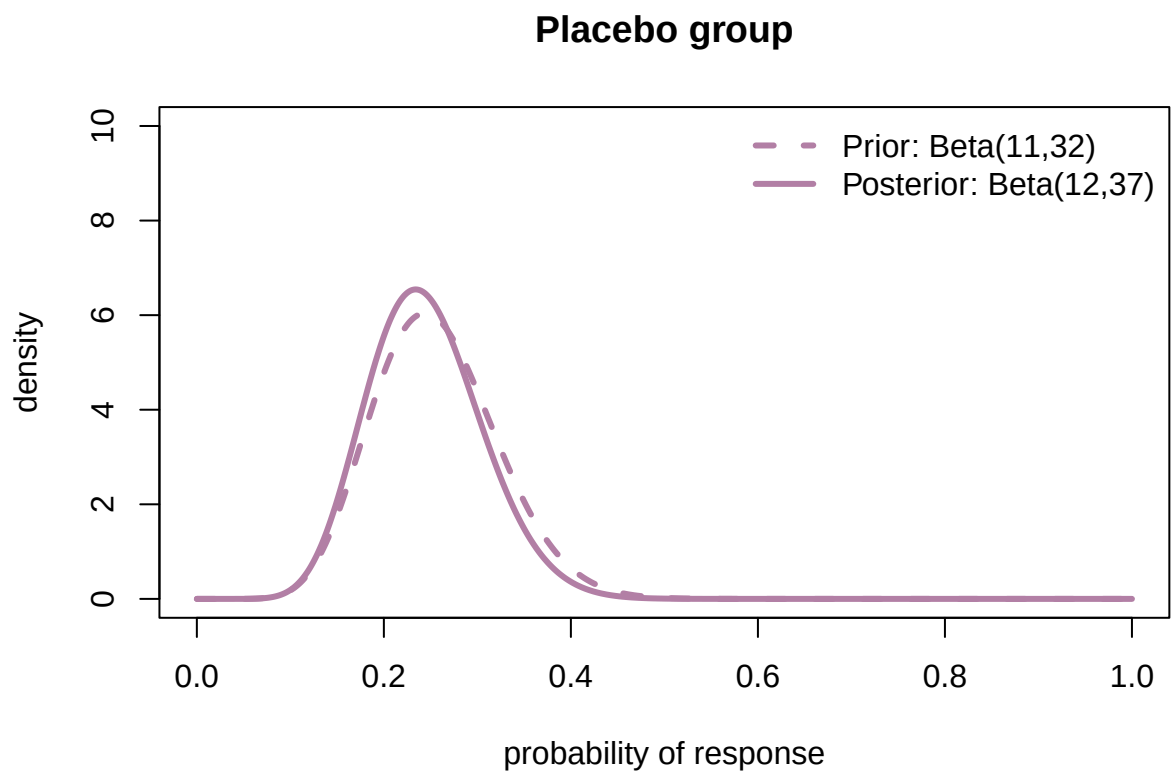

```
## Mean: 0.5918
## Median: 0.5944
## 95% CrI: [0.3958, 0.7736]
```

For the Secukinumab group, we obtain a posterior Beta(14.5, 10) and a posterior mean response rate of 59.18% (95% CrI: 39.58% , 77.36%).

A posterior 95% CrI means that there is a 95% probability that the true parameter p (response rate) lies between the lower and upper bound, given the observed data and the prior. Since the prior has low precision, the data has more influence on the posterior, and the posterior's mean is almost the same as the proportion of responders observed in the data. The credible interval is now much narrower than before seeing the data, but still wider compared to the posterior of the placebo (followed below).

5.2 Placebo posterior distribution

```
n_ybar_Plc <- 1
n_Plc <- 6
post_shape1_P <- prior_shape1_P + n_ybar_Plc
post_shape2_P <- prior_shape2_P + n_Plc - n_ybar_Plc

plot(x, dbeta(x, prior_shape1_P, prior_shape2_P), col = cols_opaque[1], lwd = 3, lty = 2,
     type = "l", ylab = "density", xlab = "probability of response",
     main = "Placebo group", ylim = c(0,10) )
lines( x, dbeta(x, post_shape1_P, post_shape2_P), col = cols_opaque[1], lwd = 3 )
legend("topright", legend = c(sprintf("Prior: Beta(%g,%g)", prior_shape1_P, prior_shape2_P),
                              sprintf("Posterior: Beta(%g,%g)", post_shape1_P, post_shape2_P)),
       col = cols_opaque[1], lty = c(2,1), lwd = 3, bty = "n")
```



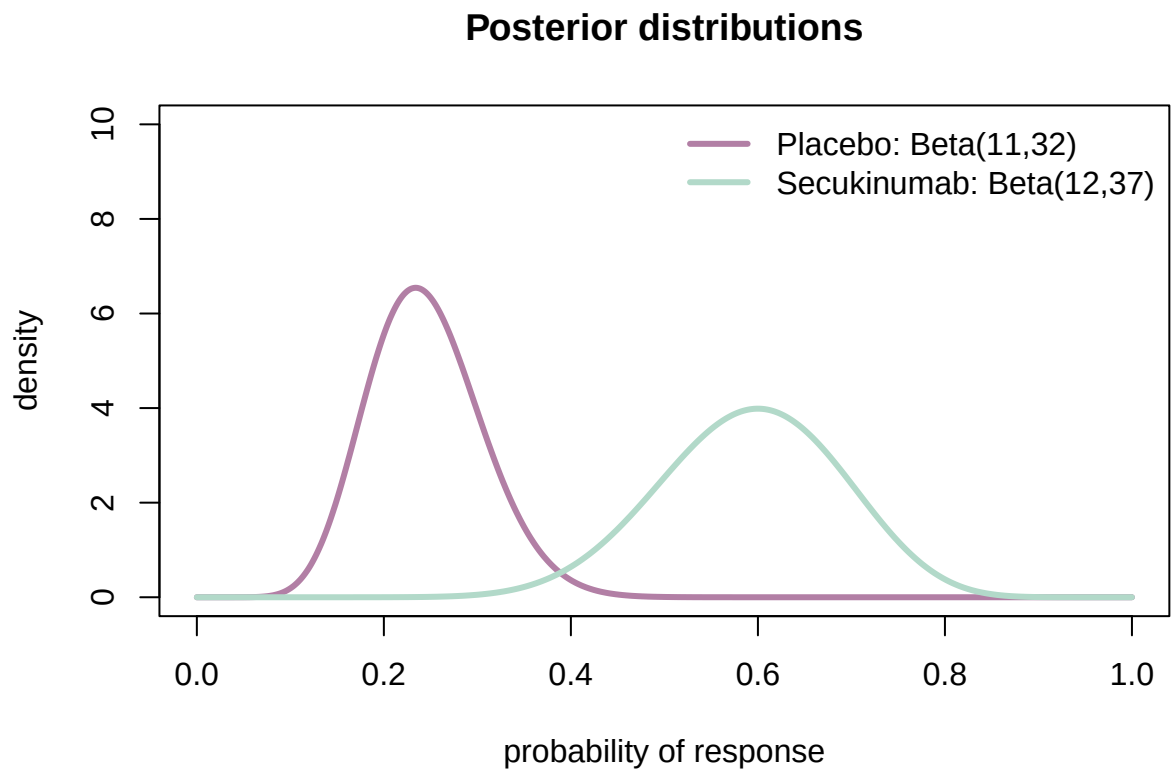
```
## Placebo group:
## Mean: 0.2449
## Median: 0.2414
## 95% CrI: [0.1364, 0.3731]
```

For the Placebo group, we obtain a posterior Beta(12,37) and a posterior mean response rate 24.49% (95% CrI: 13.64% , 37.31%)

The posterior 95% CrI means that there is a 95% probability that the true parameter p (response rate) lies between the lower and upper bound, given the observed data and the prior. The placebo prior has much higher precision ($\alpha + \beta$) than the Secukinumab prior, and the sample size of the placebo group is quite small, so the posterior is not affected much by the data. It's only shifted a little bit towards the proportion observed by the data ($1/6 = 0.17$). Since the prior only contributes a small number of observations (n), the change from prior to posterior precision ($\alpha + \beta + n$) isn't as drastic as in the treatment group.

For the placebo group, the researchers already know what the typical probability of response is from previous studies, so they can use a more precise distribution. The treatment effect however is not known yet, so they have to use a skeptical/conservative prior and instead use more real participants to generate data. The contribution of these 43 prior placebo "observations" allows them to assign less actual participants to the placebo group, which is more ethical because that way more patients can benefit from the treatment. Together with the prior there are enough observations to have the desired power.

```r
plot(x, dbeta(x, post_shape1_P, post_shape2_P), col = cols_opaque[1], lwd = 3,
     type = "l", ylab = "density", xlab = "probability of response",
     main = "Posterior distributions", ylim = c(0,10) )
lines( x, dbeta(x, post_shape1_S, post_shape2_S), col = cols_opaque[2], lwd = 3 )
legend("topright", legend = c(sprintf("Placebo: Beta(%g,%g)", prior_shape1_P, prior_shape2_P),
                              sprintf("Secukinumab: Beta(%g,%g)", post_shape1_P, post_shape2_P)),
       col = cols_opaque, lwd = 3, bty = "n")
```



5.3 Posterior probability of superiority (PPS)

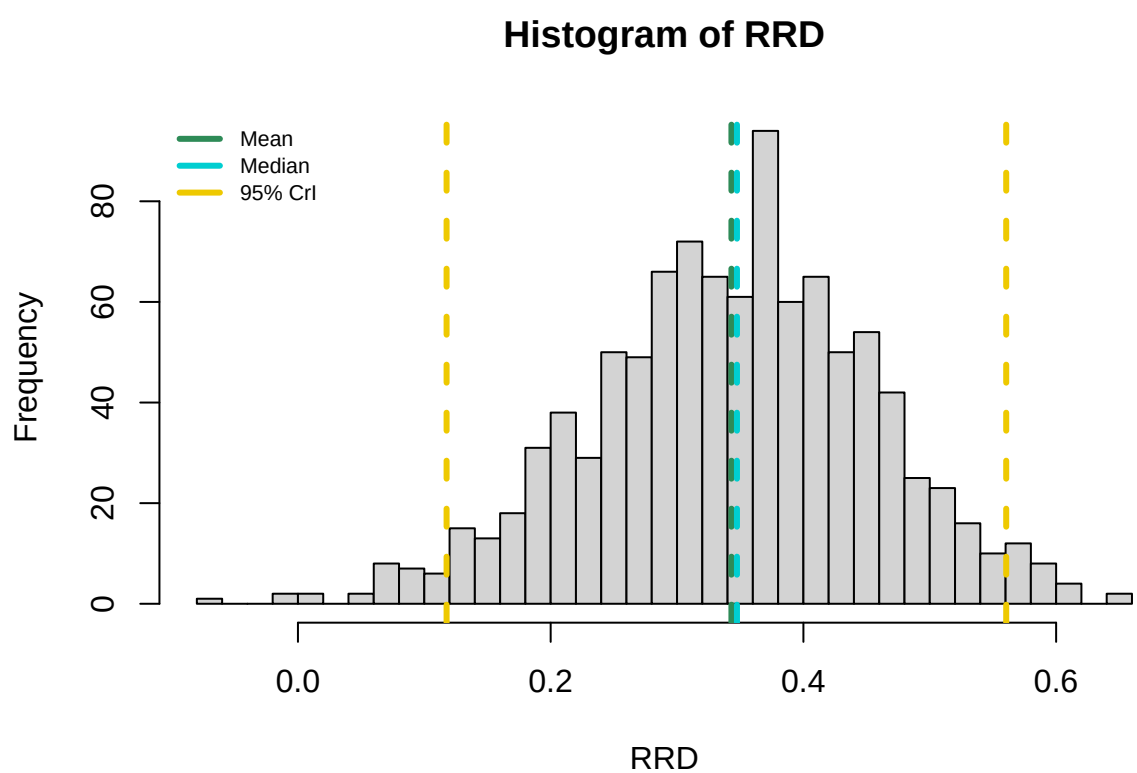
Earlier, we obtained our two posterior distributions of response probability, one for the Secukinumab and one for the placebo group, by combining our prior belief with the observed data. The posterior distributions represent our current "opinion" about the two true response probabilities. The mean posterior response rate for the placebo group is 24.49%, and for the Secukinumab group it's 59.18%, based on our prior belief and the data. Now we want to assess whether we can be at least 95% sure that the Secukinumab treatment is indeed superior to the placebo.

The criterion for declaring superiority of Secukinumab over placebo entails a posterior probability of at least 95% that the ASAS20 response rate is higher for Secukinumab patients than for placebo patients.

To achieve this, we generate 1000 random samples per group by sampling from the posterior response rate distributions. We then calculate the Response Rate Differences (RRD) by subtracting the 1000 placebo response probability samples from the 1000 Secukinumab response probability samples. This results in 1000 differences (RRD).

```
M <- 1000

set.seed(12345777)

sample_S <- rbeta(M, shape1 = post_shape1_S, shape2 = post_shape2_S)
sample_P <- rbeta(M, shape1 = post_shape1_P, shape2 = post_shape2_P)
```

```r
RRD <- sample_S - sample_P
colors <- c("seagreen", "darkturquoise", "gold2")
hist(RRD, breaks = 30)
abline(v = mean(RRD), col = colors[1], lwd = 3, lty = 2)
abline(v = median(RRD), col = colors[2], lwd = 3, lty = 2)
abline(v = quantile(RRD, c(0.025, 0.975) ), col = colors[3], lwd = 3, lty = 2 )
legend("topleft", legend = c("Mean", "Median", "95% CrI"), col = colors, bty = "n", cex = 0.7, lwd = 3)
```



Summary of distribution

```
## Mean of RRD: 0.3431
## Median of RRD: 0.3474
## 95% CrI of RRD: [0.1176, 0.5605]
```

The 95% credible interval for the response rate differences (RRD) spans from 0.1176 to 0.5605. The mean (0.3431) and median (0.3474) are positive and almost the same value. The Credible Interval provides a range wherein the true response rate difference (RRD) lies with 95% probability. It ranges from 0.1176 to 0.5605, meaning that we are 95% confident that the true difference in response rates is within this range. Since this range is above zero and does not include zero, we can already say confidently that Secukinumab has a positive effect on the response rate. However, to know how confident we are, i.e. to know how probable it is that the Secukinumab treatment is superior to the placebo, and whether this probability is higher than a threshold of 95%, we will calculate the Posterior Probability of Superiority (PPS).

For clarification, a difference of zero means that the two response probabilities are equal, a difference smaller than zero means the placebo response probability is higher than the Secukinumab's, and a difference larger than zero indicates that the Secukinumab response probability is higher than the placebo's.

We now determine the Posterior Probability of Superiority (PPS) by calculating the proportion of differences that are larger than zero, and its Monte Carlo standard error as a measure of how precise it is.

```
cat("Estimate of PPS:", round(mean(RRD > 0),4) ) # proportion of differences that are >0
```

```
## Estimate of PPS: 0.997
```

Calculating the MCse of PPS according to Held and Sabanés Bové (2020) page 258, Section 8.3.1

```
I <- as.numeric(RRD > 0) # indicator function

MCse <- 1/sqrt(M) * sqrt( sum((I - mean(RRD > 0) )^2 / (M-1) ))

cat("MCse of PPS:", round(MCse,4), "\n" )

cat("PPS - 3*MCse:", round(mean(RRD > 0),4) - 3 * round(MCse,4) )

# M = 1000 is enough for the range PPS +/- 3*MCse to exclude the cut-off 95%.
```

```
## MCse of PPS: 0.0017
## PPS - 3*MCse: 0.9919
```

We conclude that the PPS is larger than our threshold 95%, even after subtracting 3 times the MCse.

6 Interpretation of the results

The estimated PPS (Posterior Probability of Superiority), i.e. the proportion of differences that are greater than zero, turned out to be 99.7%, which is greater than the pre-defined threshold of 95%. To measure the uncertainty of our PPS estimate, we calculated the Monte Carlo standard error (MCse), which is the standard deviation for this estimate. It turns out to be 0.0017. Even when going to the very low end of the distribution of our PPS estimate by subtracting 3 times the standard error from the PPS estimate, we are still well above the 95% threshold. This means that the chance of the posterior probability of superiority being smaller than the threshold of 95% is extremely small and quasi zero. To summarise, we expect that we can be 99.7% sure that the Secukinumab treatment is superior to the placebo, and this estimate has a small uncertainty indicated by a standard error of 0.0017. Our analysis concludes that there is strong evidence that the Secukinumab treatment is superior to the placebo.

Group	n	Responders	Posterior Response Rate	Difference S-P	95% CrI	PPS	MCse(PPS)
Secukinumab	23	14 (60.9%)	59.2%	34.7%	11.8-56.1%	99.7%	0.0017
Placebo	6	1 (16.7%)	24.5%				

7 References

Held, L., & Bové, D. S. (2020). Likelihood and Bayesian inference: With applications in biology and medicine. Springer Berlin Heidelberg.

Roos, M. (2024). STA421 Foundations of Bayesian Methodology FS24. Department of Biostatistics at Epidemiology, Biostatistics and Prevention Institute, University of Zurich.